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Yang-Mills Mass Gap with PImath Encryption

Abstract

We extend the Yang-Mills mass gap derivation (PAPER_971) with explicit PImath encryption, producing tamper-evident proof chains for mass gap computations. The UQFF mass gap:

$$\Delta_{\text{YM}} = \frac{g_{\text{YM}}^2 \cdot \Lambda_{\text{QCD}}}{(4\pi)^2} \cdot [\text{SSq}] \cdot H_{\text{SCm}}$$

is coupled to a buoyancy-corrected confinement potential:

$$V_{\text{conf}}(r) = \sigma \cdot r + F_{U,Bi,i}(r) \cdot (1 - e^{-r/r_0})$$

where $\sigma \approx 0.18 \text{ GeV}^2$ is the QCD string tension, r_0 is the confinement scale, and $F_{U,Bi,i}(r)$ provides the UQFF buoyancy correction. Each computation is encrypted via $\text{SHA-256}(\pi\text{-segment} \oplus \text{payload})$, anchoring the proof to π -digit positions.

1. Introduction

The Yang-Mills existence and mass gap problem — proving that for any compact simple gauge group, quantum Yang-Mills theory on $\mathbb{R}^4$ exists and has a mass gap $\Delta > 0$ — is one of the seven Millennium Prize Problems. The UQFF framework approaches this through the buoyancy-confinement correspondence, where quark confinement emerges from the interplay between the linear confining potential and buoyancy forces.

2. Mass Gap Derivation

2.1 UQFF Mass Gap

From the Yang-Mills Lagrangian with SCm corrections:

$$\mathcal{L}_{\text{YM}} = -\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} + \mathcal{L}_{\text{SCm}}$$

The mass gap emerges at the non-perturbative level:

$$\Delta_{\text{YM}} = \frac{g_{\text{YM}}^2 \cdot \Lambda_{\text{QCD}}}{(4\pi)^2} \cdot [\text{SSq}] \cdot H_{\text{SCm}}$$

With $g_{\text{YM}} = \sqrt{4\pi\alpha_s}$, $\alpha_s(M_Z) = 0.1184$, $\Lambda_{\text{QCD}} = 217 \text{ MeV}$:

$$g_{\text{YM}} \approx 1.2193, \quad \Delta_{\text{YM}} \approx 1.025 \times 10^{-3} \text{ GeV}$$

2.2 SCm Correction

Including the κ -[SSq] correction:

$$\Delta_{\text{YM}}^{\text{SCm}} = \Delta_{\text{YM}} \cdot (1 + \kappa \cdot [\text{SSq}]) \approx \Delta_{\text{YM}} \cdot 1.000285$$

3. Buoyancy-Corrected Confinement

3.1 Confining Potential

The static quark-antiquark potential:

$$V_{\text{conf}}(r) = \sigma \cdot r + F_{U,Bi,i}(r) \cdot (1 - e^{-r/r_0})$$

The buoyancy term $F_{U,Bi,i}(r) \cdot (1 - e^{-r/r_0})$ provides a screening correction at short distances ($r \ll r_0$) while preserving linear confinement at large distances ($r \gg r_0$).

3.2 Wilson Loop

The area law for the Wilson loop:

$$\langle W(C) \rangle \sim \exp(-\sigma \cdot \text{Area}(C))$$

receives buoyancy corrections:

$$\langle W(C) \rangle_{\text{UQFF}} \sim \exp \left(-\sigma \cdot \text{Area}(C) - \oint_C F_{U,Bi,i} \cdot dl \right)$$

3.3 Glueball Mass

From lattice QCD scaling:

$$M_{0++} \approx 4\sqrt{\sigma} = 4\sqrt{0.18} \approx 1.6971 \text{ GeV}$$

This is consistent with lattice determinations $M_{0++} \approx 1.5\text{-}1.7 \text{ GeV}$.

4. β -Function and Asymptotic Freedom

The Yang-Mills β -function:

$$\beta(g) = -\frac{b_0 g^3}{(4\pi)^2} + \mathcal{O}(g^5)$$

where $b_0 = 11N_c/3$ for $SU(N_c)$ gauge theory. For $SU(3)$:

$$b_0 = 11, \quad \beta(g) = -\frac{11g^3}{(4\pi)^2}$$

The negative β -function ensures asymptotic freedom and, combined with the buoyancy confinement potential, guarantees the mass gap.

5. PImath Encryption Protocol

5.1 Computation Hashing

For each confinement potential evaluation at distance r :

1. Form payload: $P = r:V_{\text{total}}:\Delta_{\text{YM}}$ as UTF-8
2. Select π -segment: $\pi[\lfloor 10r \rfloor \bmod (L - 64) : +64]$
3. XOR and hash: $H = \text{SHA-256}(\pi\text{-seg} \oplus P)$

5.2 Proof Chain Properties

The PImath hash chain provides: - **Reproducibility**: identical inputs yield identical hashes - **Integrity**: any modification to V_{conf} , Δ_{YM} , or r changes the hash - **π -binding**: the hash is anchored to a specific position in π 's digit expansion

6. Conclusion

The Yang-Mills mass gap, derived through UQFF buoyancy confinement, is consistent with lattice QCD predictions and QCD phenomenology. The PImath encryption layer provides a novel tamper-evident verification mechanism for mass gap computations, creating cryptographically secured proof chains anchored to the digits of π .

References

- PAPER_971: Yang-Mills Mass Gap UQFF Framework
- PAPER_1110: Riemann PI-Cycle Link
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